

heart itself, and also known as coronary artery disease which is the main cause of heart attacks; Congenital Heart Disease, affecting those born with heart disease and in whom the condition slowly progresses towards advanced heart failure later in life; Arrhythmias, or abnormal heart rhythms which acutely or over time lead to damage of the heart muscle; and Valvular Heart Disease, where disease of the heart valve/s also progressively lead to heart failure. High blood pressure, coronary artery disease, diabetes, obesity, and tobacco use, continue to represent the main risk factors. Typical symptoms of heart failure include a combination shortness of breath on exertion, palpitations, swelling of the lower limbs, abdominal swelling, limited exercise capacity, difficulty lying flat, persistent cough, and weight gain from water retention. Continued symptoms and deterioration in heart function, despite optimal medical therapy is when patients should be

referred to a transplant centre for advanced heart failure management, and early assessment of candidacy for heart transplantation.

Currently, the global registry data reveals survival after heart transplantation to be 85 – 90% at one year, 75% at 3 years, 65% at 5 years, 50% at 10 years and 25% at 20 years. These are excellent outcomes where most recipients lead normal lives, return to work should they choose to, and infrequently require hospital admission. Notable recent innovations in the surgical management of end-stage heart failure include the use of donor hearts from individuals following donation after circulatory death (DCD), the improving outcomes from implantable artificial heart pumps in the form of ventricular assist devices (VAD), and the emerging prospect of xenotransplantation. Together with expert colleagues, I was able to perform and introduce the first global series of DCD heart transplants in 2014 at St Vincent's Hospital,

Sydney, Australia. It is now increasingly adopted in Europe and the USA, as a result of which 30% more heart transplant are being performed in participating centres. Whilst VAD therapy has not reached equipoise with heart transplantation in relation to long-term outcome and quality of life, it offers real and evidenced therapy for patients who are ineligible for transplantation, either as a temporary bridging therapy before transplantation, or as destination or life-long treatment. Finally, just a week short of the 2023 World Heart Day, the team at the University of Maryland Medical Center (UNMC) performed their second xeno pig-heart transplant. In 2022, the first such recipient survived until 60 days. Given the excellent initial reports on the recipient's recovery after this second attempt, we cautiously anticipate better outcome and a new ground-breaking paradigm in the evolution of heart transplantation.

ORGAN DONATION SAVES LIVES. DONATE LIVES THROUGH TRANSPLANTATION

Dr. Swaminathan Sambandam, Senior Consultant & Lead, Multi-Organ Transplant (Liver, Kidney, Pancreas & Small Intestine), Kauvery Hospitals, Tamil Nadu assures that organ transplantation is carried out as per the guidelines of the apex regulatory body National Organ and Tissue Transplant Organization (NOTTO).

He was the first to perform Robotic Kidney Transplantation in Tamil Nadu, besides achieving numerous other milestones.

What is organ failure?

Any organ can begin to fail in performing its expected function. Such failure can deteriorate to such a degree that normal life cannot be sustained without external clinical intervention or life support. For example, the liver performs hundreds of functions necessary to support life, including filtering toxins out of the blood. Kidney balances fluid and electrolytes in the body and remove wastes from the blood into the urine.

How has technology changed in organ transplantation, and will lab-grown organs be a reality?

Minimally invasive surgery (MIS), using laparoscopic or robotic techniques, has revolutionized the field of transplantation. MIS has many advantages over conventional open surgery that include faster post-operative recovery, fewer pulmonary complications and an excellent cosmetic outcome.

Across the world, extensive research is being undertaken to find alternatives including lab-grown organs and harvesting organs from genetically altered pigs. Science is making remarkable progress in both but, as of now, the main source of organs for transplantation is donation, from live or deceased donors. Lab grown organs or bioprinted organoids are incomplete, lacking some of the vasculature and infrastructure of organs. To successfully grow organs and tissues there must be a biologically compatible

3D scaffold which contains all the biochemical messages in the correct configuration to trigger the formation of the desired organ or tissue.

Vigorous campaigns for cadaver donation are necessary to meet the substantial need for organ transplantation. How much ground has India covered in launching and sustaining them?

We need to encourage and sustain year-round, and country-wide, efforts to promote organ donation. One such initiative is the Indian organ donation week organized from 20th to 27th of November when events are conducted to create awareness among the public to understand the importance of cadaver organ donation. The percentage of such organ donation in a country like Spain is 34%, US 26%, and UK 13% whereas India has only 0.8%. A lack of awareness about organ donation is the main factor responsible for the inadequate from the general population. Religious beliefs and customs stoke the fear that the body handed over towards harvest of organs would not be treated with respect and dignity. To overcome the current shortage of donated cadavers, efforts need to be made to favourably influence the mindset of the wider Indian society towards organ donation. A conversation about donation of organs after death needs to become an integral part of the preparation towards care at the end-of-life.



Doctor, can we have your assurance to the reader, that organ transplantation the best alternative to any other?

Transplantation is the most appropriate treatment, the gold standard, for end stage organ failure. That can increase the longevity and quality of life. It will enable individuals who have reached that stage in their illness to live healthier and longer lives; the science, and the evidence, are equally in support of that. Society and science together need to sincerely work towards making that possible, with a clear and conscious regard and respect for donors and recipients. It is the ultimate human experience, the gift of organ donation and the ability to give life through transplantation. Life begins again.